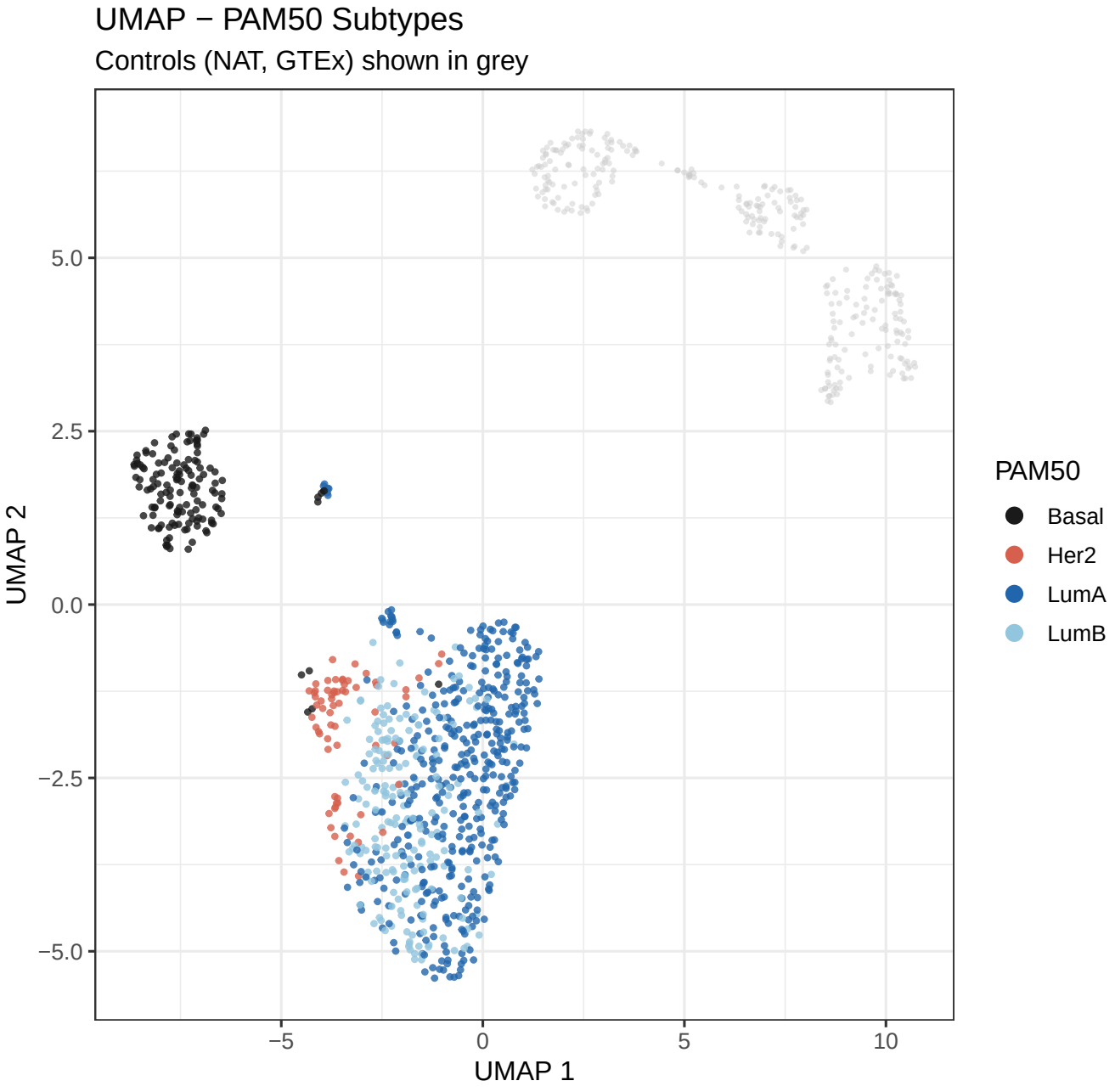
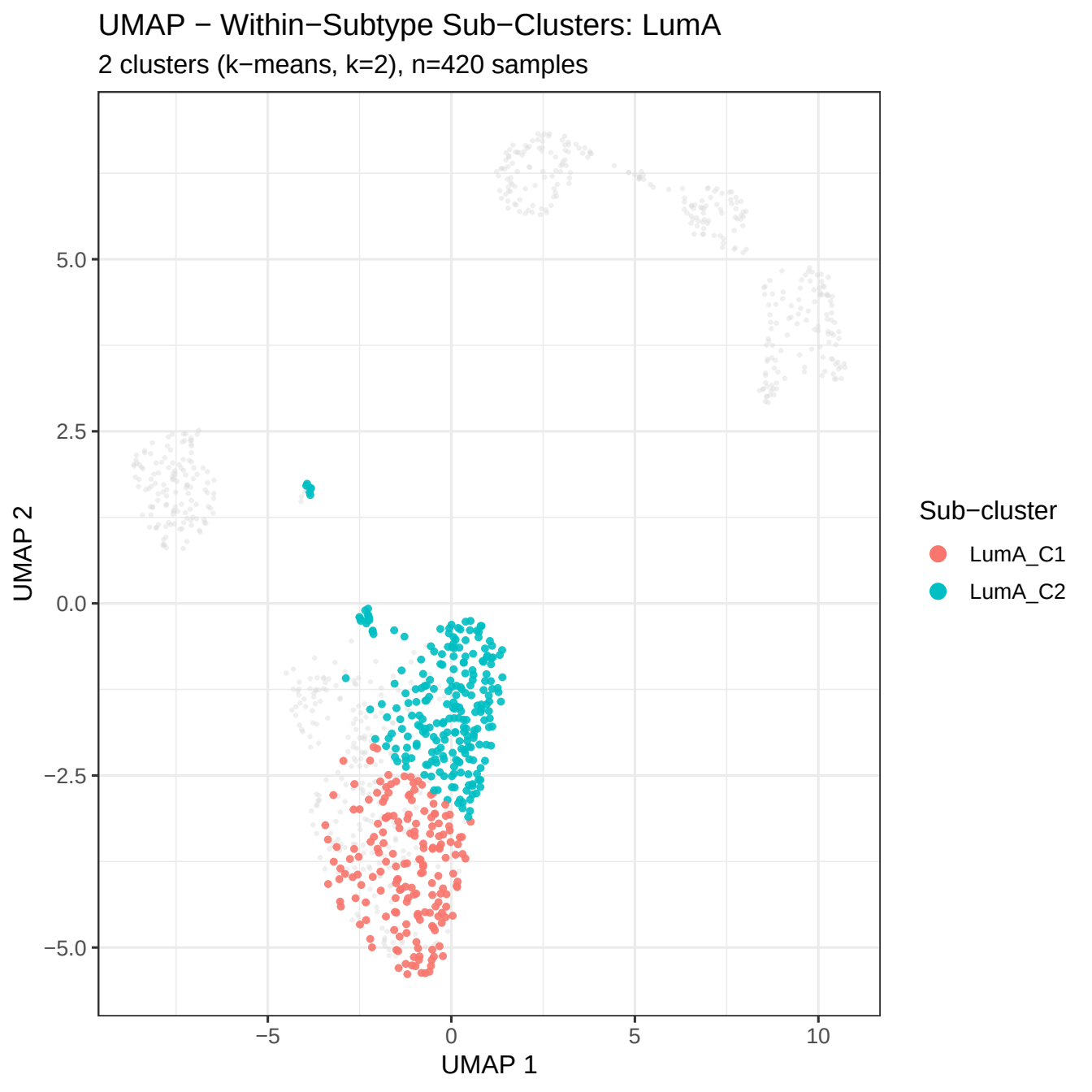
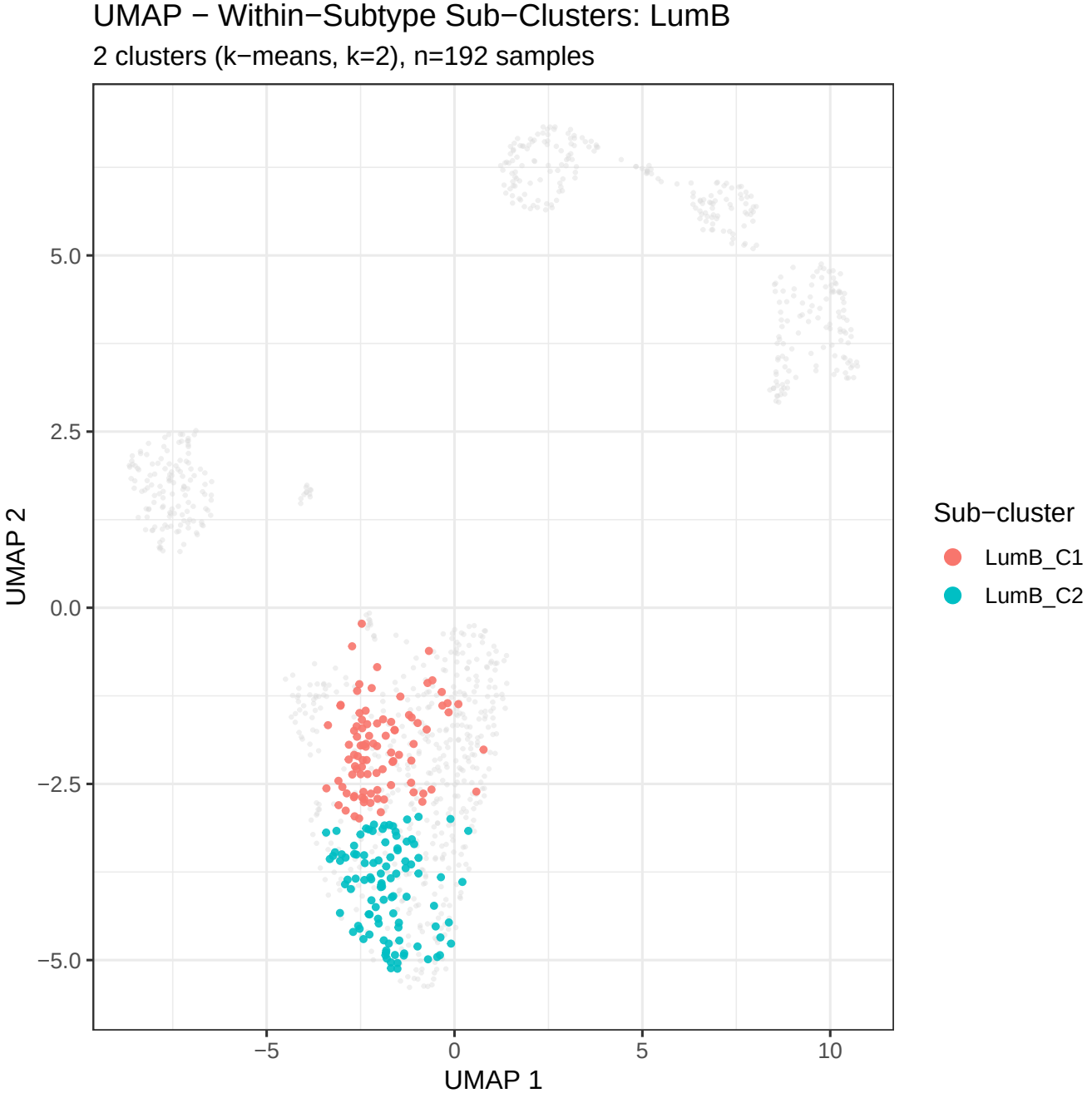
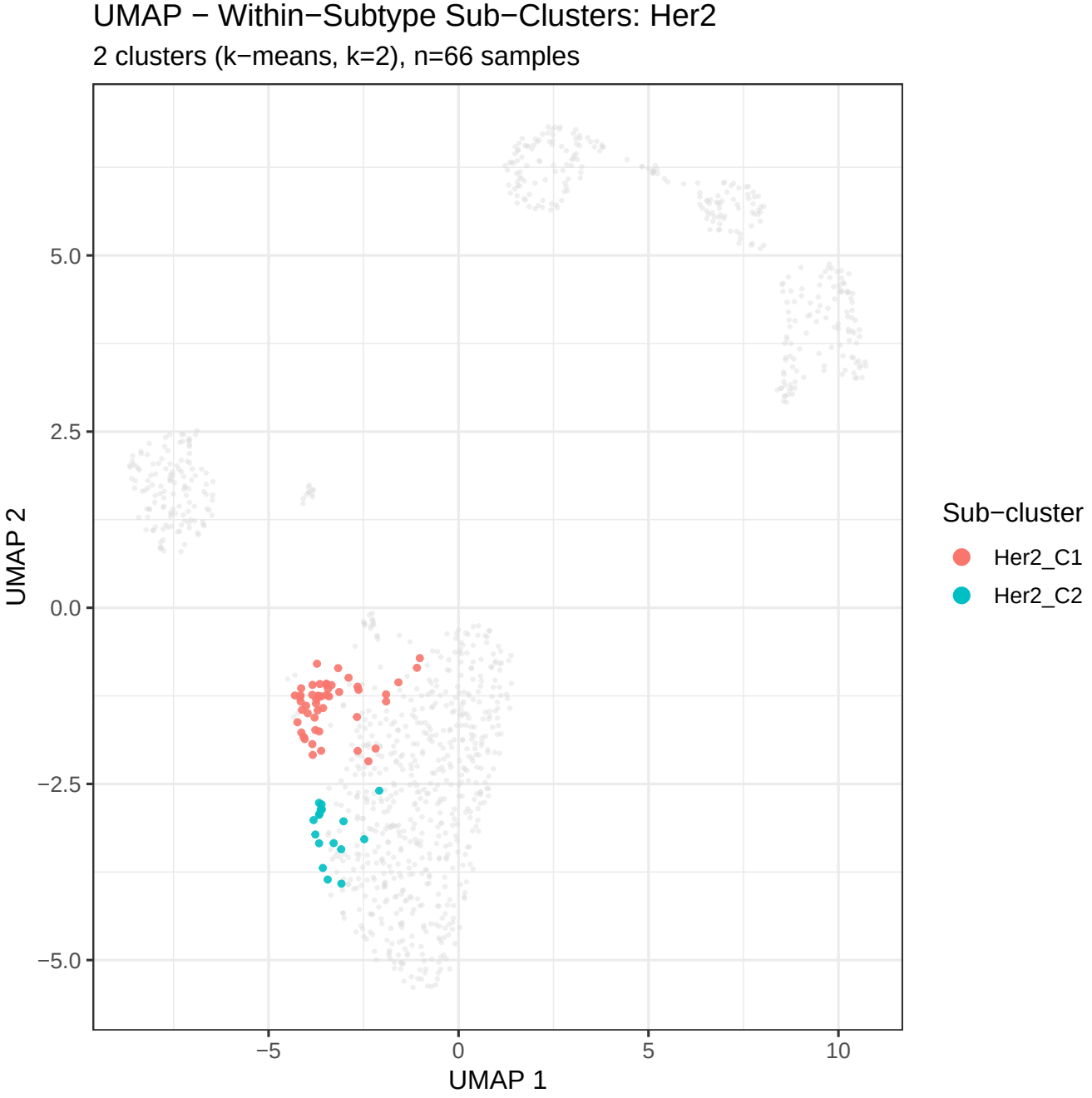
UMAP − Sample Groups
Tumor (PAM50−classified) · NAT · GTEx Healthy Breast
5.0
2.5
0.0
−2.5
−5.0
UMAP 2
−5
0
5
10
UMAP 1
Group
GTEx
NAT
Tumor
UMAP (5000 HVGs, 50 PCs, n_neighbors=30). n = 1109 samples.

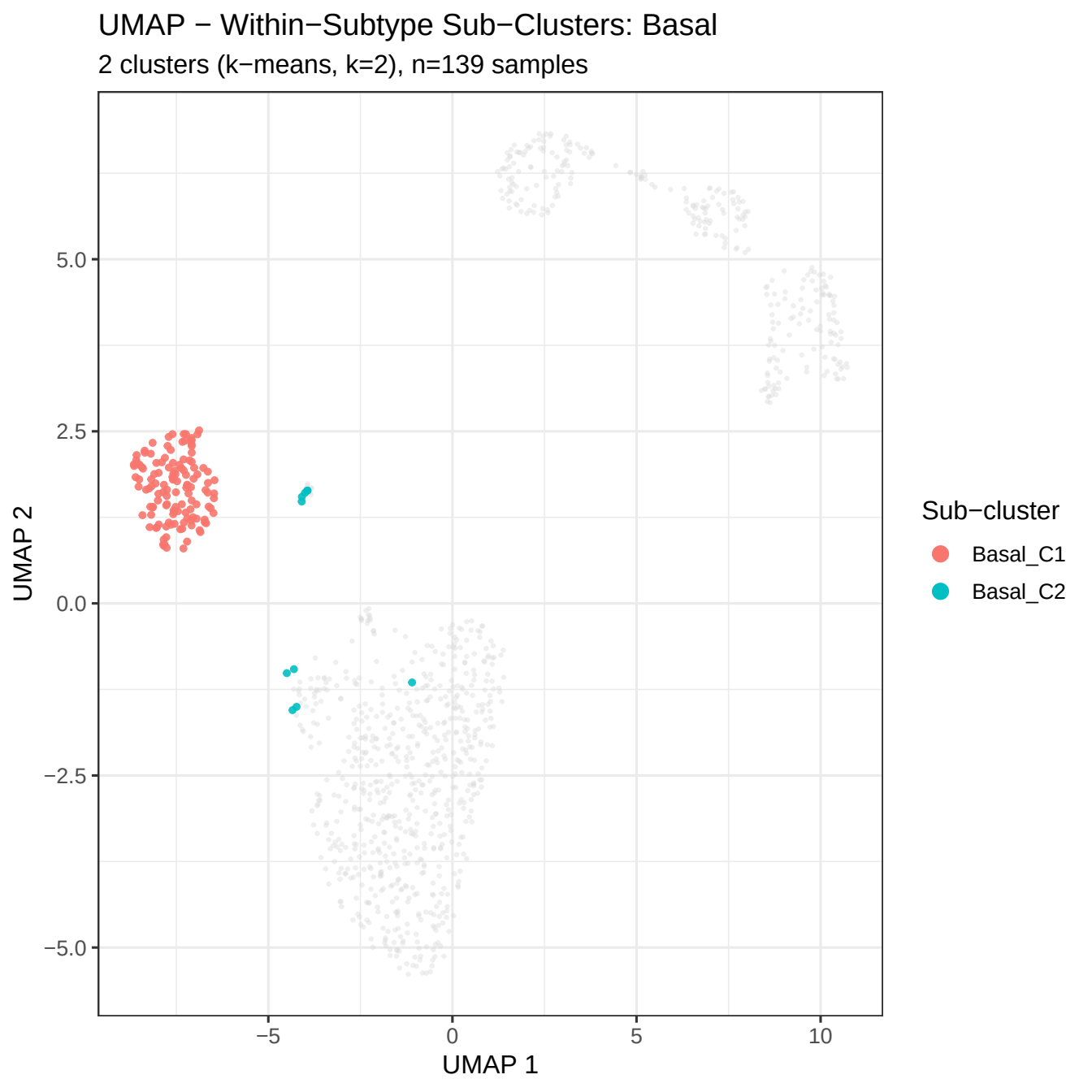


PAM50 subtypes: LumA n=420, LumB n=192, Her2 n=66, Basal n=139.


UMAP − Within−Subtype Sub−Clusters: LumA
2 clusters (k−means, k=2), n=420 samples
UMAP 2
UMAP 1
5.0
2.5
0.0
−2.5
−5.0
−5
0
5
10
Sub−cluster
LumA_C1
LumA_C2


UMAP − Within−Subtype Sub−Clusters: LumB
2 clusters (k−means, k=2), n=192 samples
5.0
2.5
0.0
−2.5
−5.0
UMAP 2
−5
0
5
10
UMAP 1
Sub−cluster
LumB_C1
LumB_C2


UMAP − Within−Subtype Sub−Clusters: Her2
2 clusters (k−means, k=2), n=66 samples
5.0
2.5
0.0
−2.5
−5.0
UMAP 2
−5
0
5
10
UMAP 1
Sub−cluster
Her2_C1
Her2_C2


UMAP − Within−Subtype Sub−Clusters: Basal
2 clusters (k−means, k=2), n=139 samples
UMAP 2
5.0
2.5
0.0
−2.5
−5.0
−5
0
5
10
UMAP 1
Sub−cluster
Basal_C1
Basal_C2